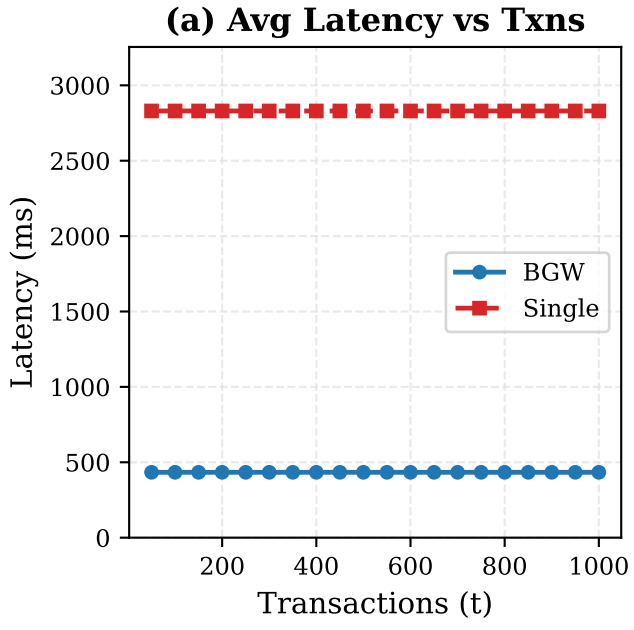
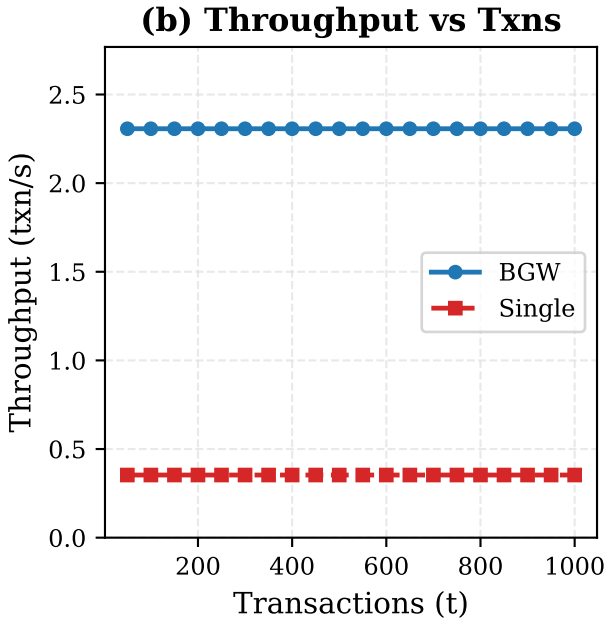
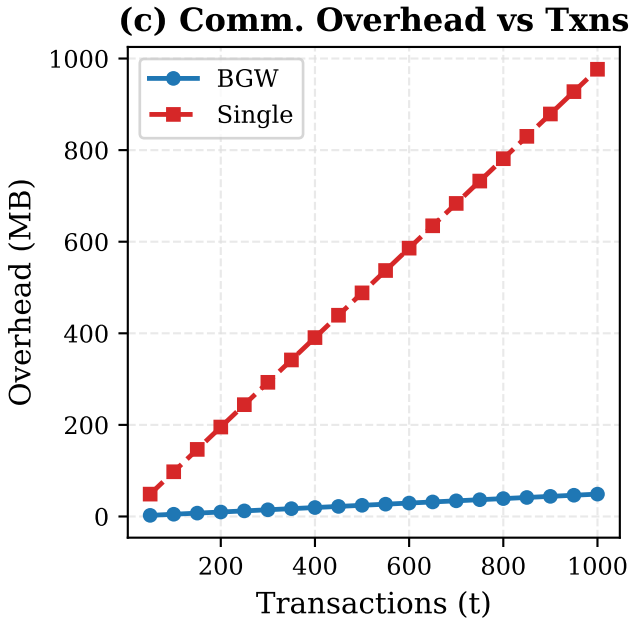
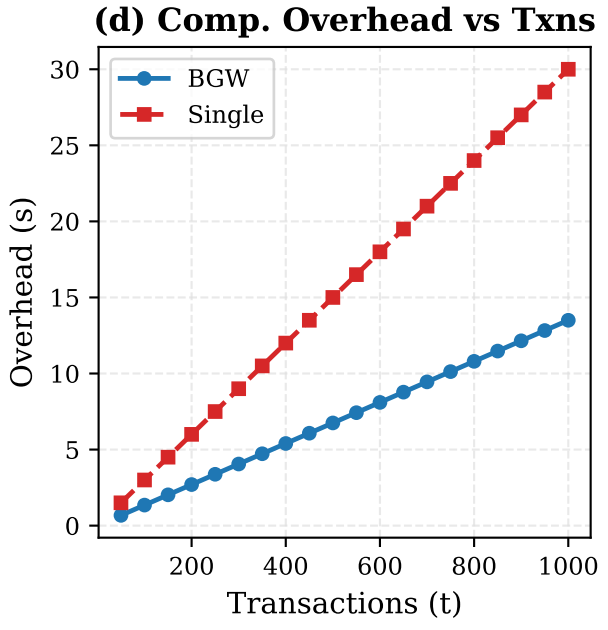
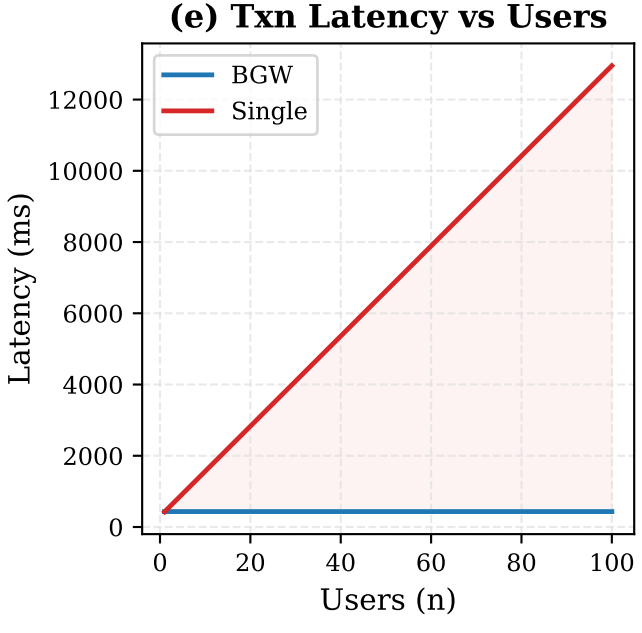
Performance Analysis: BGW Broadcast Encryption vs Single Encryption (Blockchain E-Health Access Control System)

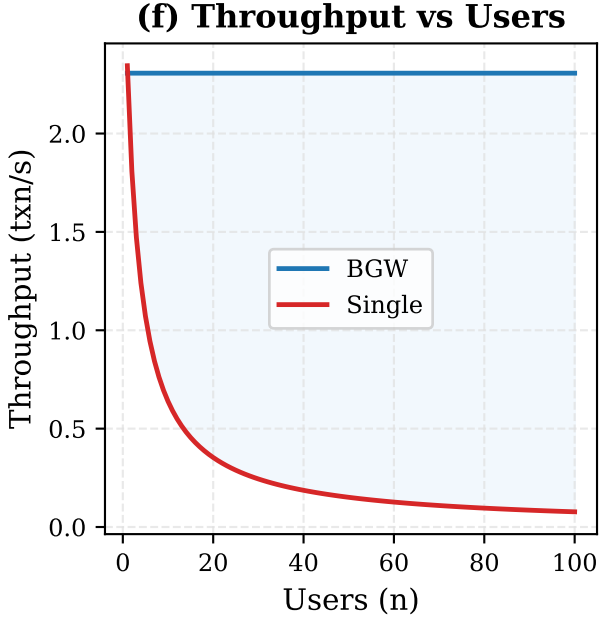


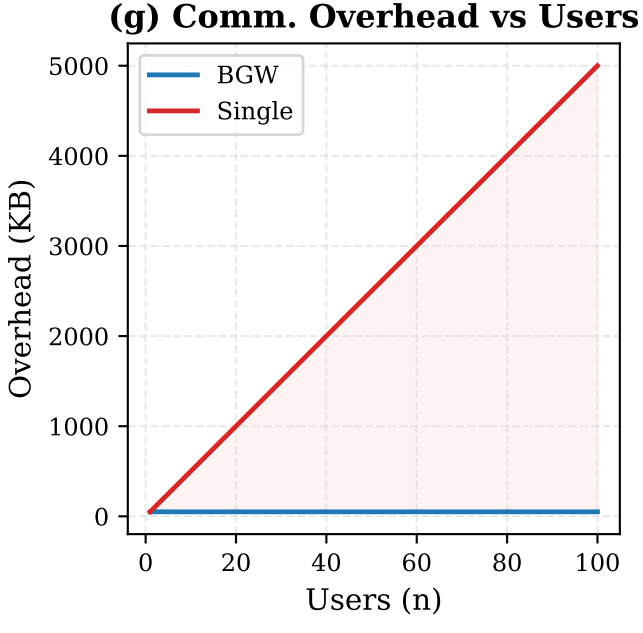












Theoretical Simulation

```
T_AES        = 1.5 ms
T_BGW_hdr    = 12.0 ms
T_BGW_setup  = 38.0 ms
T_IPFS       = 120.0 ms
T_BC         = 300.0 ms
T_key_send   = 5.0 ms
File Size D = 50 KB
n (for vs t)= 20

BGW: Encrypt once, 1 header
Single: Encrypt n times
```